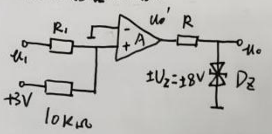
Homework 12.

1. 单限比较器电路：

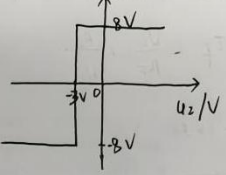


当 $u_I > U_T$ 时，$U_P > U_N$，即 $u_0' = +U_{OM}$，$u_0 = U_{OH} = +U_Z$.
当 $U_I < U_T$ 时，$U_P < U_N$，即 $U_0' = -U_{OM}$，$U_0 = U_{OL} = -U_Z$.
$u_0 = \pm U_Z = \pm 8V$

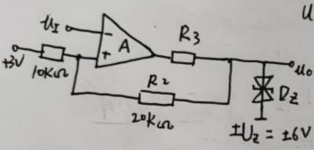
令 $u_P = u_N$ 可得 $u_I = -3V = U_T$.

$$U_P = \frac{R_1}{R_1+R_1} U_I + \frac{R_1}{R_1+R_1} \times 3V$$

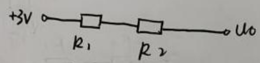
即电压传输特性如右图所示：



2. 滞回比较器.



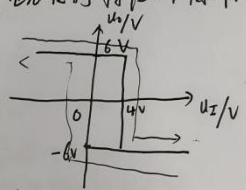
$u_I < -U_T$，$u_N < u_P$，$u_0 = +U_Z$，$u_P = +U_T$
限幅：$u_0 = \pm U_Z = \pm 6V$.



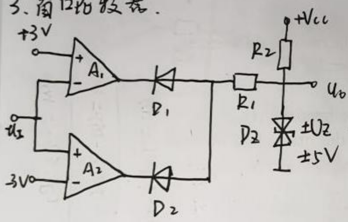
$$U_P = \frac{R_1}{R_1+R_2} u_0 + \frac{R_2}{R_1+R_2} U_{REF} = U_N = U_I$$

得 $U_{T1} = 0V$，$U_{T2} = 4V$.

即电压传输特性如下图所示：



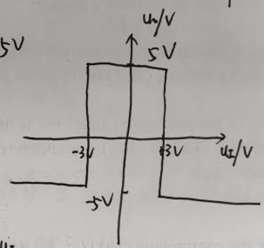
3. 窗口比较器.



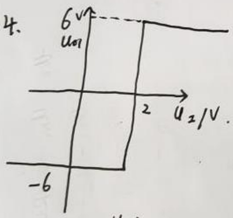
$u_0 = \pm U_Z = \pm 5V$

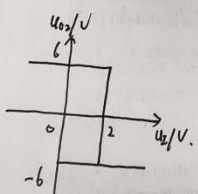
$\pm U_T = \pm 3V$

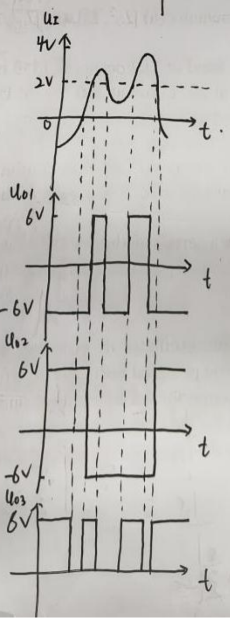
传输特性：

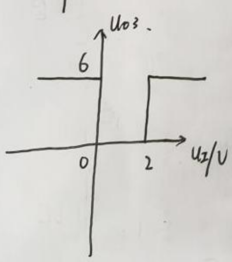


4.





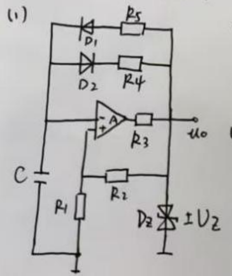




Homework 12.

5、$R_1=R_2=20k\Omega$, $R_3=4K\Omega$, $R_4=20K\Omega$, $R_5=40K\Omega$, $C=0.05\mu F$, $\pm U_Z=\pm10V$.

(1)

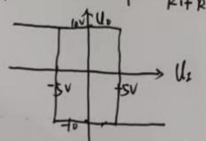


输出电压 $u_O=\pm10V$. 振荡周期 $T\approx(R_4+R_5)\cdot C\ln(1+\frac{2R_1}{R_2})$
$=3.30ms$

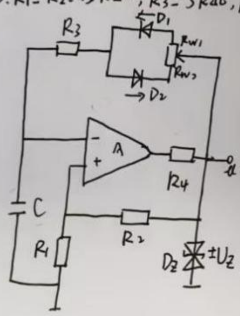
占空比: $q=\frac{T_1}{T}=\frac{R_5}{R_4+R_5}=\frac{2}{3}=0.67$

(2) $\pm U_T=\pm\frac{R_1}{R_1+R_2}\cdot U_Z$ $\quad U_P=U_N$ $\quad U_P=\frac{R_2}{R_1+R_2}u_O$

代入得 $U_T=\pm5V$



6. $R_1=R_2=25K\Omega$, $R_3=5K\Omega$, $R_W=100K\Omega$, $C=0.1\mu F$, $\pm U_Z=\pm8V$.



(1) 输出电压 $u_O=\pm8V$, 振荡周期.

$T\approx(R_W+2R_3)C\ln(1+2\frac{R_1}{R_2})$
$=12.1ms$.

振荡频率: $f=\frac{1}{T}\approx83Hz$

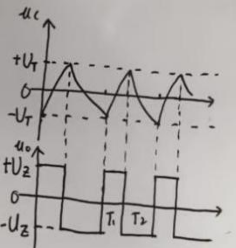
(2) 当 R_{W1} 为最小值代入, 可得 q 最小值.

$q_{min}=\frac{T_1}{T}\approx\frac{R_{W1}+R_3}{R_W+2R_3}=\frac{5}{100+10}\approx0.045$

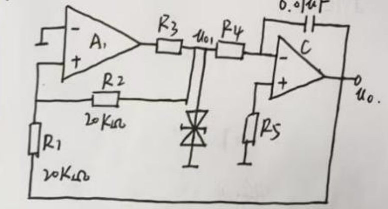
将 R_{W1} 为最大值 $100k\Omega$ 代入, 可得 q 最大值.

$q_{max}=\frac{T_1}{T}\approx\frac{R_{W1}+R_3}{R_W+2R_3}=\frac{100+5}{100+10}\approx0.95$

即占空比 $T_1/T\approx0.45\sim0.95$.



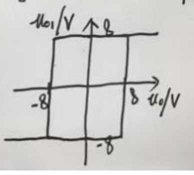
7.



(1). A_1: 滞回比较器 $\quad A_2$: 积分运算电路.

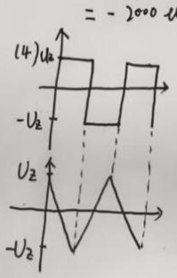
(2). $u_{P1}=\frac{R_1}{R_1+R_2}\cdot u_{O1}+\frac{R_2}{R_1+R_2}\cdot u_O=\frac{1}{2}(u_{O1}+u_O)=u_{N1}=0$

得: $\pm U_T=\pm8V$; 电压传输特性曲线:



(3) $u_O=-\frac{1}{R_4C}u_{O1}(t_2-t_1)+u_O(t_1)$
$=-2000u_{O1}(t_2-t_1)+u_O(t_1)$

(4)



(5) 提高振荡频率可以减小 R_4, C, R_1 或增加 R_2.

$T=\frac{4R_1R_4C}{R_2}$